MECH 325 Reading Guide #5
Fasteners, Linear Drives, & Springs

1 Overview

In this module, we will be covering springs, threaded fasteners, power and ball screws.

As always, the readings are divided into two categories:

- **Required**: the primary source of material for the Quiz. Each student is expected to complete all the required readings for the entire module.
- ***Beneficial:*** *additional analyses, derivations, explanations and examples to provide in-depth understanding of the course material. The beneficial readings will not appear directly on the RAP quiz but will help develop the more complete understanding of course concepts necessary for the tutorials, exercises, exams, and design projects.*

All readings are drawn from the course texts: Budynas, R.G. and Nisbett, J.K., *Shigley's Mechanical Engineering Design*, McGraw-Hill. Page numbers: **2024 Release page numbers in bold**, 11th Edition in regular and *10th & 9th Edition in italics*. The other text is Mott, R.L., Vavrek, E.M. and Wang, J. *Machine Elements in Mechanical Design*, 6th Edition, Pearson Education.

2 Readings

2.1 Introduction to Threaded Fasteners

Required (Shigley): Section 8-1 **(pp. 434-38,** pp. 422-26**,** *pp. 402-5, 410-3)*. This section covers the terminology and standards for screw threads.

Required (Shigley): Section 8-2 **(pp. 438-41,** pp. 426-29, *pp. 406-9, 414-17*). This section introduces the power screw mechanism and provides the relationships involving power, torque and force.

Or Required (Mott): Section 17-2 This section introduces the power screw mechanism and provides the relationships involving power, torque and force.

Beneficial*: Section 8-2* **(pp. 441-45,** pp. 429-32, *pp. 410-14, 418-22). The remaining pages in section 8-2 provide details on thread strength as well as an example for sizing a power screw. Mott 17-2 example is also a sizing example.*

Required (Shigley): Sections 8-3 and 8-4 (**pp. 446-49**, *pp. 414-19, 422-7*): these sections introduce various types of threaded fasteners and their uses as well as the important concepts of fastener preload and stiffness.

Beneficial*: Section 8-5* (**pp. 449-55,** pp. 437-43, *pp. 419-24, 425-30): this section describes the effect of bolt on the fastened members in terms of the stiffness of those members.*

Required (Shigley): Section 8-6 **(pp. 455-58,** pp. 443-46, *pp. 424-7, 432-5*): this section discusses bolt strength and the associated standards. Take a look over the tables but do not memorize any of the information.

Or Required (Mott): Section 19-2 this section discusses bolt strength and the associated standards. Take a look over the tables but do not memorize any of the information.

Required (Shigley): Sections 8-7 to 8-9 (**pp. 458-67,** pp. 446-54, *pp. 427-34, 435-42)*: these sections combine the concepts above into statistical analysis of tension joints. Section 8-7 discusses the external applied load, 8-8 describes the relationship between bolt torque and preload, and 8-9 considers the entire

joint. *(The examples are beneficial but not required reading.)*

***Beneficial (Mott)**: Section 19-4* (pp. **696-98**)*: this section describes the relationship of bolt preload to input torque. It describes a variety of methods to measure clamping forces.*

***Beneficial (Mott)**: Section 19-6* (pp. **700**)*: this section describes the failure mode where the threads are stripped. This topic is not discussed in Shigley.*

Required (Shigley): Section 8-11 (**pp. 468-71,** pp. 456-60, *pp. 436-40, 444-8*): this section covers tension-loaded bolted joints subject to fatigue. (*Example 8-5 is beneficial but not required reading.*)

Required (Shigley): Section 8-12 (**pp. 474-80,** pp. 463-69, *pp. 443-9, 451-7*): this section examines bolted joints loaded in shear. Pay particular attention to the section on eccentric loading (pp. **478-80**, 467-9, *455-7, 439-41*). (*The examples are beneficial but not required reading.*)

***Beneficial (Mott)**: Section 20-3* (pp. **710-12**)*: this section also describes eccentric loading of bolts. (The example is very similar to Shigley Example 8-5).*

2.2 Introduction to Ball Screws

Required (Mott): Section 17-3 (pp. **649-651**): this section gives an introduction to ball screws and the relationships of life, torque and power.

Required (Mott): Section 17-4 (pp. **652**): this section discusses column bucking and critical speed.

Additional Information on Power Screws

***Beneficial**: Thomson Linear Motion (thomsonlinear.com). Thomson is one of the leading suppliers of ball screws. The ball screw catalogue can be found online and also in the background materials section on the Canvas site. https://www.thomsonlinear.com/downloads/screws/Ball_Screws_Splines_cten.pdf*

Skim the first 8-10 pages to get a sense of how these mechanisms operate

Engineering Information: Read pages 139-142, 146-149 (imperial) or pages 150-153 and 155-158 (metric). The equations and Figures provide selection criteria for life, bucking, and speed considerations.

2.3 Introduction to Linear Bearings

The topic of linear bearings (often called ball bushings or rail guides) is not covered in any of the prescribed textbooks. However, these types of bearings are used for numerous applications. There are several industrial supplies of linear bearings. As Thomson Linear Motion (thomsonlinear.com) is a leader in this area, we will use the Round Rail Linear Guide and Components Catalogue which can be found on Canvas or online as our technical resource. https://www.thomsonlinear.com/en/support/docs-linear-bearings-guides

***Beneficial**: Go to Canvas (or Thomson website) to access the Thomson RoundRail Linear Guides Catalogue.*

Skim the first 15 pages of the catalogue to get a sense of how these mechanisms operate and the variety of available options.

Engineering Information: Skim the first few pages in the Engineering Section (pages 252-56). Read pages 257-61. The equations and Figures provide selection criteria for load and life considerations.

2.4 Introduction to Springs

Required (Mott): Section 18-1 – 18-4 (pp. **657-667**): this section gives an introduction to helical springs and compression springs.

Or Required (Shigley): Sections 10-1 to 10-4 (pp. **526-29,** 510-3, *518-21*) gives an introduction to helical springs and compression springs.

Beneficial*: Shigley Section 10-5 provides information on stability and Section 10-6 addresses spring materials. Similar stability information is provided in Mott Section 18-5*

Required: (Mott): Section 18-6 (pp. 670-73) Skim the example to see how springs are designed.

Beneficial (Shigley): *Section 10-7 (pp.* ***535-40,*** *520-4, 528-30) the material covers spring design for static loading in more detail. (Example 10- 2 is beneficial, but not required reading.)*

Beneficial (Shigley)*: Section 10-8 (*pp. **542-3,** 526-8**,** *534-6) discusses the critical frequencies in helical springs.*

Required (Shigley): Section 10-9 (pp. **543-7**, 528-31, *536-7*) covers spring design for fatigue loading. *(Example 10-4 and the extended example of Section 10-10 are beneficial but not required.)*

Required (Shigley): Section 10-11 (pp. **550-7,** 534-42, *542-5*) introduces extension springs.

Beneficial (Mott)*: Sections 18-7 and 18-8 (pp. 677-682) Provides an overview of extension and torsional springs*

Or Beneficial (Shigley) Sections 10-11 (pp. 550-57, 534-42, *542-5)* ***and 10-12 (****pp.* **557-9,** 542-3, *550-1) introduces extension and torsion springs.*